

5. Model Evaluation



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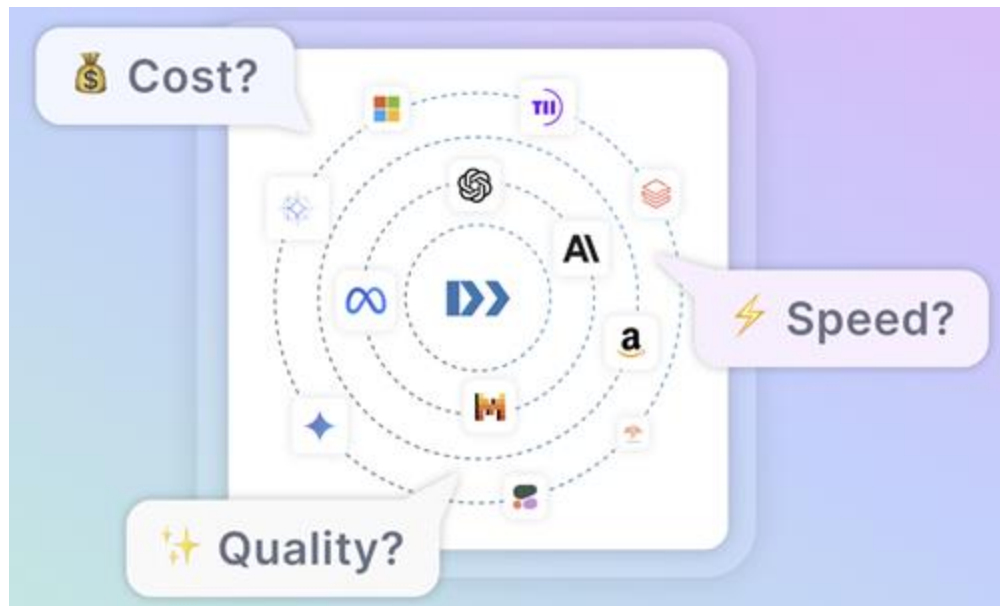
5. Model Evaluation

1. Model Evaluation
2. Evaluation for Automatic Speech Recognition (ASR)
3. Evaluation for Text-to-Speech (TTS) & Voice Cloning
4. Challenges and Future Directions

5.1. Model Evaluation

- Assess how well a model performs on **unseen data**
- Measure progress and compare different models or algorithms

=> A poorly evaluated model can lead to usability issues, mistrust in AI systems, or even harm in critical



Dataset

- Train: Seen
- Dev / Validation: Seen
 - Parameter Optimization
 - Error Analysis
- Test: Unseen

Importance of Model Evaluation

- Direct impact on user experience (e.g., intelligibility of TTS or accuracy of ASR)
- Application domains are sensitive to errors, such as healthcare (speech profiling for diseases) or customer service (ASR for call centers).
- Real-time applications demand not just accuracy but also efficiency and robustness.

Discussion

- Why might traditional metrics like accuracy or F1-score be insufficient for evaluating speech models?
- Can you think of any real-world applications where evaluating speech models poorly could have significant consequences?

Discussion

- Loss of Temporal Information
 - Speech is inherently sequential and time-dependent.
 - Accuracy and F1-score do not capture alignment errors, which are crucial in Automatic Speech Recognition (ASR) and Text-to-Speech (TTS).

Common metrics for different tasks

- **STT (ASR):** Word Error Rate (WER), Syllable Error Rate (SylER), Character Error Rate (CER), Phoneme Error Rate (PER)
- **TTS:** Mean Opinion Score (MOS), Intelligibility (WER), Mel Cepstral Distortion (MCD)
- **Voice Cloning:** Similarity Score, WER, Mean Opinion Score (MOS), Signal-to-Noise Ratio (SNR)
- **Voice Conversion:** Similarity Score, WER
- **Speaker Verification:** EER (Equal Error Rate)
- **Speaker Identification:** Accuracy, Top-k accuracy, Confusion Matrix

5.2. Evaluation for ASR Task

- Automatic Speech Recognition (ASR) refers to converting spoken language into text.
- Accurate ASR is critical for applications like virtual assistants, transcription services, and voice-controlled systems.
- Challenges in ASR Evaluation:
 - Variability in accents, dialects, and background noise.
 - Vocabulary size and domain specificity.
 - Balancing accuracy and real-time performance.

Word Error Rate (WER)

Definition and Significance

- Word Error Rate (WER) is the most commonly used metric to evaluate ASR systems.
- It measures the percentage of words incorrectly predicted by the ASR system compared to a ground truth reference.

Calculation Process: $WER = (S + D + I) / N$

where:

- S = Substitutions (words replaced incorrectly)
- D = Deletions (words omitted)
- I = Insertions (extra words added)
- N = Total number of words in the reference

Word Error Rate (WER)

Reference Sentence:



Tôi thích ăn phở gà.

ASR Output:



Tôi yêu ăn gà

bò.

Reference	ASR Output	Error Type
Tôi /toj/	Tôi	 Correct
Thích /tʰik/	yêu	 Substitutions
ăn	ăn	 Correct
phở	—	 Deletions
gà	gà	 Correct
—	bò	 Insertions

Word Error Rate (WER)

WER Calculation:

- Substitutions (S): 1 (“thích” → “yêu”)
- Deletions (D): 1 (“phở” is missing)
- Insertions (I): 1 (“bò” were added)
- Total words in reference (N): 5

$$\text{WER} = (S + D + I) / N = (1 + 1 + 1) / 5 = 0.6$$

Word Error Rate (WER)

Variants of WER

- Character Error Rate (CER): Measures errors at the character level, useful for languages with complex scripts.
- Syllable Error Rate (SylER): Measures errors at the syllable level, useful for languages with complex scripts.
- Sentence Error Rate (SER): Measures the percentage of sentences with at least one error.

Word Error Rate (WER)

Strengths and Limitations

- Strengths: Widely understood, easy to calculate, and directly related to ASR quality.
- Limitations:
 - Does not consider semantic correctness.
 - Penalizes minor errors equally to major ones.
 - Sensitive to out-of-vocabulary words and proper nouns.

Other ASR Metrics

Accuracy and Word Accuracy (WAcc)

- Accuracy = (Correctly recognized words / Total words) × 100%.
- Word Accuracy (WAcc) adjusts WER by accounting for correctly recognized words:
$$WAcc = 1 - WER$$

Word Recognition Rate (WRR)

- Measures the proportion of words recognized correctly out of the total spoken words.

Time to Transcription (Latency)

- Measures the time taken by the ASR system from receiving the audio input to generating the text output.
- Critical for real-time applications like live captioning or voice assistants.

Human vs. Machine Performance

- Human transcription is often used as a gold standard, but even human transcribers have error rates, typically around 5-6%.
- Recent ASR systems, especially those using deep learning, have approached and even matched human-level accuracy in controlled environments.
- Case Study: Google's ASR system achieved a WER of 5.1% on the Switchboard dataset, comparable to human transcription performance.

ASR Evaluation Best Practices

- **Dataset Considerations**

- Use large, diverse datasets covering multiple accents, speaking rates, and background conditions.
- Common benchmarks include LibriSpeech, Switchboard, and Common Voice.

- **Speaker Diversity and Acoustic Conditions**

- Include speakers from different age groups, genders, and regional accents.
- Evaluate performance in noisy environments, telephone channels, and varying microphone qualities.

- **Continuous Evaluation in Real-World Deployments**

- Regularly evaluate ASR systems with live data to adapt to evolving language use, slang, and domain-specific vocabulary.
- Employ user feedback and error analysis to improve system accuracy over time.

5.3. Evaluation for TTS/Voice Cloning

- **Objective** (automatic, computable)
 - Mel Cepstral Distortion (MCD)
 - Latency and real-time factor (RTF)
 - Speaker Similarity
- **Subjective** (based on human perception)
 - MOS, CMOS, SMOS
 - ABX or A/B Test
- **Intelligibility**: Word Error Rate (WER)

Perception Test

A perception test in Text-to-Speech (TTS) systems evaluates how well a synthetic voice performs in terms of its naturalness, intelligibility, and listener preferences. These tests aim to assess the quality of TTS output from the perspective of human listeners.

Common types of perception tests used in TTS evaluation:

- Mean Opinion Score (MOS)
- ABX Test
- AB Test
- Intelligibility Test
- DMOS
- CMOS / SMOS
- Degradation Category Rating (DCR)
- Preference Test

Goals of Perception Testing in TTS:

- Benchmarking: Compare the system against competitors or human recordings.
- Fine-tuning: Identify areas for improvement, such as pronunciation, intonation, or fluency.
- User acceptance: Ensure the TTS system meets the expectations of its target

Mean Opinion Score (MOS): Difference aspects

- Standard human evaluation method (subjective)
- Listeners rate audio on a **scale from 1 to 5** based on **naturalness, intelligibility, or any other aspects of quality.**
- Ground-truth should be included
- Common score scale:
 - 5 = Excellent
 - 4 = Good
 - 3 = Fair
 - 2 = Poor
 - 1 = Bad

Mean Opinion Score (MOS): Naturalness

Naturalness refers to how closely the synthesized speech resembles human speech in terms of sound quality, absence of synthetic artifacts, and overall pleasantness.

Xin chào, Lê Minh Nguyễn

Đổi mật khẩu

Đăng xuất

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5 điểm - Rất tự nhiên, giống như giọng người thật.

4 điểm - Tương đối tự nhiên, khá giống giọng người thật

3 điểm - Hơi tự nhiên, khá nhiều yếu tố nhân tạo

2 điểm - Kém tự nhiên, rất nhiều yếu tố nhân tạo

1 điểm - Rất kém tự nhiên, hoàn toàn nhân tạo

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Mean Opinion Score (MOS): Naturalness

How to Perform a MOS Test?

Prepare the Test Materials

- Select a representative dataset of text prompts to synthesize. These should include diverse sentence structures, word complexities, and prosody variations.
- Generate audio samples using the TTS system being tested.

Design the Test Setup

- Choose a rating scale (typically 1-5). You can also use finer scales (e.g., 1-10) or anchor-based rating (e.g., comparing against a reference).
- Provide clear instructions to the participants on how to rate the samples.

Recruit Participants

- Use human evaluators who are fluent in the target language and familiar with the nuances of natural speech.
- Ensure a diverse pool of evaluators to reduce biases (e.g., native vs. non-native speakers, gender balance).

Conduct the Test

- Randomize the order of the audio samples to prevent bias.
- Use a blind test setup, where listeners are not aware of which system generated the audio.
- Provide headphones or a controlled environment to ensure audio quality consistency.
- Ask participants to rate each sample based on specific criteria (e.g., naturalness).

Collect Data

- Gather the ratings from all participants for all audio samples.
- If applicable, collect additional qualitative feedback or notes.

MOS: Perceptive Results

Calculate Results

Aggregate Scores: For each audio sample, compute the mean score by averaging the ratings across all participants.

- Example: If 10 participants rate a sample as [4, 4, 5, 3, 4, 5, 4, 5, 3, 4], the mean score is: $MOS = (4 + 4 + 5 + 3 + 4 + 5 + 4 + 5 + 3 + 4) / 10 = 4.1$

Calculate Overall MOS: Average the mean scores of all the audio samples to get the overall MOS for the TTS system.

Statistical Analysis:

- Compute the standard deviation (SD) to understand the variability in scores.
- Use confidence intervals (CI) to assess the reliability of the MOS. A common confidence level is 95%.

Different metrics for MOS

- **MOS**: "How good is this voice?"
- **DMOS**: "How degraded is this compared to the original?"
- **CMOS**: "Which one is better?"
- **SMOS**: "Does it sound like this speaker / emotion / style?"

Metric	Full Name	Purpose	Type
MOS	Mean Opinion Score	General naturalness/quality	Absolute
DMOS	Degradation MOS	Perceived degradation vs reference	Relative
CMOS	Comparison MOS	Preference between two samples	Relative
SMOS	Style/ Speaker MOS	Style similarity or speaker identity	Absolute

DMOS (Degradation MOS)

- DMOS evaluates how much worse a **degraded signal** is compared to a reference (clean) version.
- Listeners compare original vs. distorted speech, and rate the perceived degradation.
- **Scale:**
 - Usually also from 1 (highly degraded) to 5 (no degradation).
- **Use cases:**
 - Used in speech enhancement, denoising, codec evaluation, or low-bitrate synthesis.
 - Common in telecom, VoIP, or **noisy TTS** outputs.

CMOS (Comparative MOS)

- Listeners **compare** two TTS samples (A vs B) and judge which one sounds better.
- Instead of absolute scoring, CMOS captures **relative preference**.
- **Scale:** From -3 to +3:
 - +3: A is much better than B
 - 0: No preference
 - -3: B is much better than A
- **Use cases:**
 - Directly compare two TTS systems (e.g., your model vs baseline)
 - Useful in A/B testing for model iterations

SMOS – Style MOS (or Speaker MOS)

- **Style MOS**

- Evaluates how well the TTS conveys speaking style, such as emotion, expressiveness, or intonation.
- Listeners rate: “Does this voice sound expressive, angry, happy, neutral...?”

- **Speaker MOS**

- Evaluates speaker similarity in voice cloning.
- Question to listener: “Does this voice sound like the target speaker?”

Intelligibility: Word Error Rate (WER)

Definition and Relevance

- **Intelligibility** measures how easily the synthesized speech can be understood.
- **Word Error Rate (WER)** is commonly used to evaluate speech recognition accuracy. For TTS, you can measure WER by having listeners transcribe the synthesized speech and comparing their transcriptions to the original text.

Intelligibility: Word Error Rate (WER)

Full-Transcript Approach:

- Provide participants with synthesized speech samples.
- Ask them to write down what they heard.
- Compare their transcription to the ground truth.
- Calculate $WER = (\text{Substitutions} + \text{Deletions} + \text{Insertions}) / (\text{Total Words}) \times 100\%$.

Semantically Unpredictable Sentences (SUS):

- SUS are sentences constructed to minimize semantic context, forcing the listener to rely purely on acoustic-phonetic cues (e.g., "The green snake liked a singing pencil.").
- Useful for isolating articulation clarity from prior language knowledge.

Intelligibility: Word Error Rate (WER)

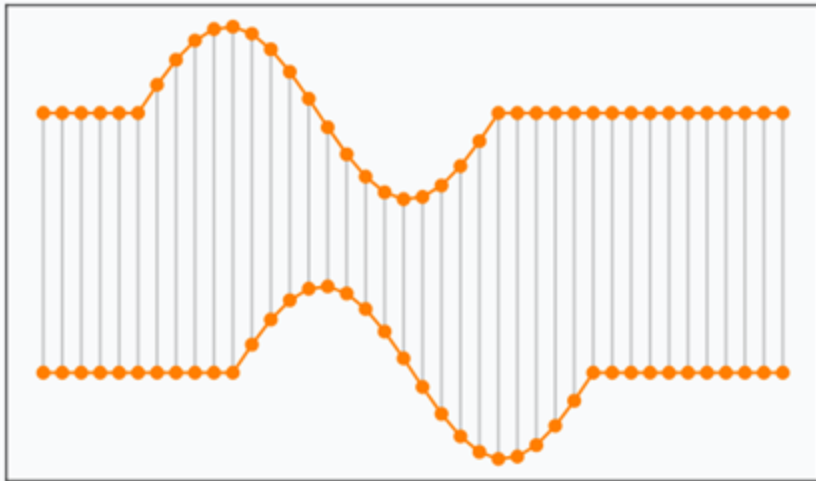
Considerations

- Listener Fatigue: Keep tests manageable in length.
- Vocabulary Familiarity: Use a representative vocabulary set relevant to the application domain if not using SUS.
- Reliability: Larger test sets and more raters improve reliability.

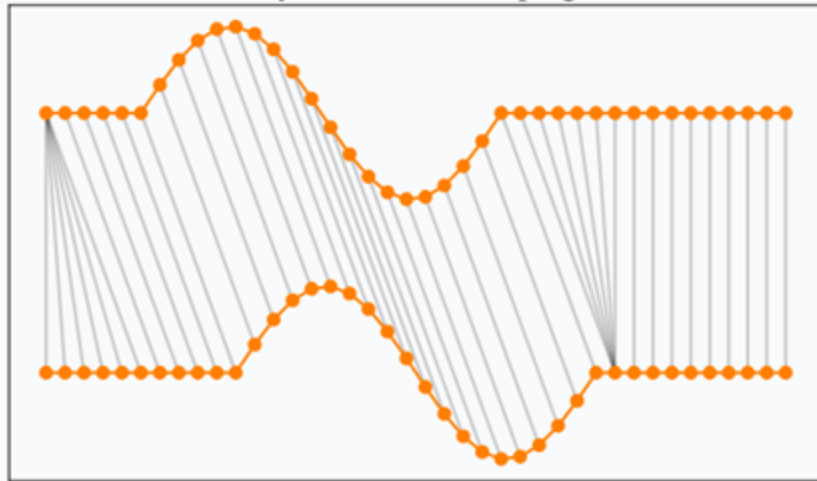
Mel Cepstral Distortion (MCD)

- Measures the distance between the mel-cepstral coefficients of real vs. synthetic audio.
 - Lower MCD = better quality
- Common in parallel TTS setups with ground truth (Euclidean distance or DTW)

Euclidean distance



Dynamic Time Warping



Latency and Real-Time Factor (RTF)

Importance in TTS Systems

- Beyond quality metrics, practical deployment requires evaluating speed and latency.
- Latency refers to the time it takes from receiving input text to generating the corresponding audio output.
- Real-Time Factor (RTF) is a common objective metric defined as:
 - **$\text{RTF} = (\text{time taken to synthesize the speech}) \div (\text{duration of the generated audio})$**
 - **If $\text{RTF} < 1$, the system is faster than real time (e.g., 0.5 RTF means 2x real-time speed).**

Latency and Real-Time Factor (RTF)

Benchmarks:

- Synthesize a representative set of text inputs of varying lengths (short sentences, paragraphs, etc.).
- Record total computation time and output speech duration.

Hardware Considerations:

- Report the hardware used (CPU, GPU, RAM) since performance can differ significantly.
- Compare different system configurations to see how they affect RTF.

Trade-offs

- Faster synthesis might reduce quality if model size or complexity is limited.
- Some methods (e.g., streaming or chunk-based synthesis) can reduce perceived latency but may affect audio continuity.

TTS vs. Voice Cloning

Aspect	TTS (Text-to-Speech) 🗣️	Voice Cloning 🗣️👤
Definition	Converts text into synthetic speech.	Reproduces a specific person's voice.
Purpose	Generate natural, intelligible speech for general use.	Generate speech that sounds like a specific speaker .
Training Data	Large datasets from multiple speakers .	Requires data from one or a few target speakers .
Customization	Uses pre-trained voices , limited speaker selection.	Can replicate any speaker's voice.
Data Required	100+ hours (traditional), 1–10 hours (modern).	Few-shot learning (minutes to hours) for cloning.
Output Quality	Highly natural , but generic voice .	Highly realistic , captures unique speaker traits .
Speaker Consistency	Voice remains generic across all outputs.	Mimics speaker's tone, style, and pitch .

Evaluation for Voice Cloning

- Ensuring high-quality, natural-sounding speech.
- Maintaining consistency and intelligibility.
- Avoiding ethical and security concerns related to synthetic voices.

Key Challenges in Voice Cloning Evaluation

- Balancing speaker similarity and intelligibility.
- Handling variability in emotions, speaking styles, and recording conditions.
- Measuring how well a cloned voice generalizes across different linguistic content.

Speaker Similarity Metrics

Subjective Evaluation: Mean Opinion Score (MOS) for Similarity

- Listeners rate how similar the cloned voice is to the target speaker on a 5-point or 7-point MOS scale.
- Example Scale:
 - 1 = Not Similar
 - 3 = Somewhat Similar
 - 5 = Very Similar
- Test setup: Randomized listening sessions where raters compare original vs. cloned speech.

Speaker Similarity Metrics

Objective Metrics: Speaker Embedding Distance

- Cosine Similarity: Measures how close the speaker embeddings of the cloned and original voices are.

$$\textit{Similarity Score} = \frac{E_{\textit{cloned}} \cdot E_{\textit{original}}}{||E_{\textit{cloned}}|| ||E_{\textit{original}}||}$$

- Higher values (closer to 1) indicate higher similarity.
- Euclidean Distance: Measures the absolute difference in feature vectors between cloned and original voices.
 - Lower values indicate greater similarity.

Speaker Similarity Test

VLSP Vietnamese Text To Speech

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Đổi mật khẩu

Đăng xuất

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☐ Giống hoàn toàn (4 điểm)

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Speaker Verification

- False Acceptance Rate (FAR): Rate of accepting an impostor (false positive)
- False Rejection Rate (FRR): Rate of rejecting a genuine speaker (false negative)
- **Equal Error Rate (EER)**: The point where $FAR = FRR$
 - Lower EER = Better system
- **Example**
 - Suppose you have:
 - 1,000 test trials (speech, claimed ID pairs)
 - 500 genuine (true speaker)
 - 500 impostor (wrong speaker)
 - SV System returns similarity scores from 0 to 1

Speaker Verification

Threshold	FAR (Impostor accepted)	FRR (Genuine rejected)
0.2	35%	5%
0.4	20%	10%
0.5	10%	10%
0.6	6%	15%
0.8	3%	28%

Speaker Identification

- Accuracy = (Number of correct predictions / Total samples) × 100%
- Example
 - You have 100 test utterances from 10 known speakers
 - The system correctly identifies 85 of them → Accuracy = 85%
- Optionally: Top-k Accuracy
 - Measures if the correct speaker is in the top k predicted candidates

Naturalness and intelligibility metrics

- **MOS for Naturalness**

- Similar to TTS evaluation, naturalness is rated on a 1-5 or 1-7 scale.
- Factors influencing naturalness:
 - Smoothness of prosody
 - Presence of robotic artifacts
 - Speech discontinuities

- **Relationship Between Intelligibility and Cloning Quality**

- A cloned voice that is highly natural but low in intelligibility is problematic.
- Ideally, WER should be close to that of the original speaker.

Ethical and Security Considerations

Detectability of Cloned Voices

- Can human listeners or AI models detect synthetic speech?
- Develop metrics to measure the likelihood of cloned speech being misidentified as human.

Misuse Risks and Watermarking Strategies

- Risks:
 - Fraud, misinformation, impersonation attacks.
- Watermarking and Traceability:
 - Embedding inaudible signals to indicate synthesized content.
 - AI-based forensic models to detect cloned voices.

Ethical Guidelines in Voice Cloning

- Ensuring consent before cloning a person's voice.
- Clearly marking AI-generated speech to avoid deception.

Challenges and Future Directions

- Cross-metric trade-offs and balancing priorities
- Subjectivity in Human Judgments
- Ethics in evaluation: Bias and fairness in speech models.
- Emerging trends in speech model evaluation (e.g., explainable AI, multimodal evaluation).